

Jordan Newman

Systems Software Engineer | Local AI, Runtimes, Operating Systems

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Willing and able to work in person in New York City

SUMMARY

Systems software engineer with more than 20 years centered on C, with C++ used across that career and periodic Python use spanning approximately 20 years. Built production native runtimes, FreeBSD kernel modules, an x86 operating-system kernel, network software, and high-scale infrastructure. Brings two-plus years of hands-on machine learning framework and local model inference work, primarily through personal projects, including LM Studio, llama.cpp, Ollama, and LM Studio Bionic. Currently building Nexus, a local-first agentic control plane with model routing, observability, secure tools, React and Tauri clients, and auditable execution. Deep experience in operating systems, software system design, technical problem-solving, communication, CPU profiling, debugging, and performance optimization.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, TypeScript, Rust, C#, assembly, Go, JavaScript, Shell
- **Local AI and ML:** LM Studio, llama.cpp, Ollama, LM Studio Bionic; familiarity with MLX, TensorFlow, Hugging Face Transformers, and CUDA
- **Runtime and Operating Systems:** FreeBSD, Linux, x86 kernel development, system calls, processes, shared memory, Unix descriptor passing, kqueue, non-blocking I/O, sendfile()
- **Debugging and Performance:** GDB, Valgrind, Linux perf, CPU profiling and optimization, event-driven systems, concurrency, zero-copy I/O
- **Application and Platform:** React, TypeScript, Tauri, Electron, SQLite, server-sent events, MCP, Docker, Kubernetes, Terraform, Azure DevOps, CI/CD
- **Networking:** TCP/IP, HTTP, DNS, sockets, BGP4, NetFlow, packet capture, CDN architecture

SELECTED AI, RUNTIME, AND SYSTEMS ENGINEERING

Nexus Agentic Operating System — Creator and Engineer

Active personal project

- Designing a local-first agent control plane that persists tasks, approvals, audit records, and layered knowledge in SQLite and the local filesystem before introducing cloud dependencies.
- Implemented deterministic model-routing profiles with health checks, circuit breakers, budget-aware provider selection, failover, and Anthropic prompt caching.
- Built a privacy-conscious model-invocation ledger tracking tokens, latency, retries, errors, and estimated cost without retaining prompt or response content.
- Developed declarative YAML, DAG-oriented multi-agent workflows with parallel execution, conditional branches, controlled loop-back behavior, and audited task-state transitions.
- Engineered default-deny tool permissions using risk classification, human approvals, workspace grants, air-gapped network controls, canonical paths, and symlink-traversal protection.
- Delivered React/Tailwind and Tauri user and developer experiences with live events, approvals, routing simulation, AI-usage statistics, keychain-backed secrets, and local process supervision.

PrimeHTTPD and Production CDN Runtime

- Architected PrimeHTTPD, a non-blocking HTTP/CDN runtime in C for FreeBSD with a single-process kqueue networking core and dedicated workers for blocking disk operations.
- Used Unix descriptor passing, mmap() shared worker state, least-busy scheduling, descriptor and gzip caches, and zero-copy sendfile() with SF_NODISKIO to protect the event loop from blocking I/O.
- Deployed the runtime across approximately 200 servers supporting more than 150,000 concurrent connections within infrastructure spanning 3,000-4,000 servers, 10 locations, and over 65 Gbps.
- Profiled and optimized CPU usage in systems software using GDB, Valgrind, Linux perf, and other native debugging and performance tools.

Operating-System and Kernel Engineering

- Built TAFOS, an educational x86 kernel in C and assembly with an MBR bootloader, protected mode, interrupt descriptor table, heap allocator, port I/O, VGA output, and GDB support.
- Developed production FreeBSD kernel modules for transparent filesystem-path virtualization, execution monitoring, system protection, telemetry, and syscall instrumentation.

Open-Source Engineering

- Contributed a merged Rust Azure Key Vault provider to cachix/secretspec, implementing authentication modes, validation, feature-gated integration tests, sovereign-cloud support, documentation, and maintainer-review fixes.

PROFESSIONAL EXPERIENCE

Advantive — Senior Development Specialist, APIs and Integrations

Remote / Tampa Bay, FL | June 2022 - Present

- Contribute to Architecture Team decisions, software-system design, engineering standards, platform direction, shared libraries, documentation, developer tooling, and AI-assisted engineering workflows.
- Implement and improve Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.

DDI System — Senior Development Specialist

Manalapan, NJ | October 2021 - July 2022

- Developed primarily C# desktop applications and UniVerse database connectivity; added OAuth 2.0 email support plus SellerCloud and product-catalog integrations.

ISPRIME — Chief Executive Officer

Weehawken, NJ | March 2018 - December 2019

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

Earlier Software Engineering

DDI System — Computer Programmer | April 2017 - March 2018

Built ERP integrations and automated order, catalog, image, safety-data, SFTP, supplier, SellerCloud, and MultiValue/UniVerse workflows.

Too Much Media LLC — Computer Programmer | January 2016 - April 2017

Developed real-time browser chat integrated with ticketing and implemented OAuth 2.0 authorization for protected customer sites.

MFCXY, Inc. — Computer Programmer | December 2014 - September 2015

Developed backend and Windows desktop-client features plus middleware for cross-platform database migration.

ISPRIME — Owner and CIO

Weehawken, NJ | January 2001 - December 2014

- Architected and operated production hosting and CDN infrastructure and personally built its HTTP serving, kernel, networking, security, monitoring, authentication, and operational software.
- Created technical training documentation, trained the majority of employees over the years, and mentored two beginners over several years into technical experts and company leaders.

ADDITIONAL EXPERIENCE

AJPM, LLC — Programmer / Owner | November 2012 - December 2013

FatWallet.com — Systems Administrator | December 2000 - June 2001

EDUCATION

Brookdale Community College — Computer Science coursework; left during the first semester to pursue ISPRIME full-time during rapid company growth.